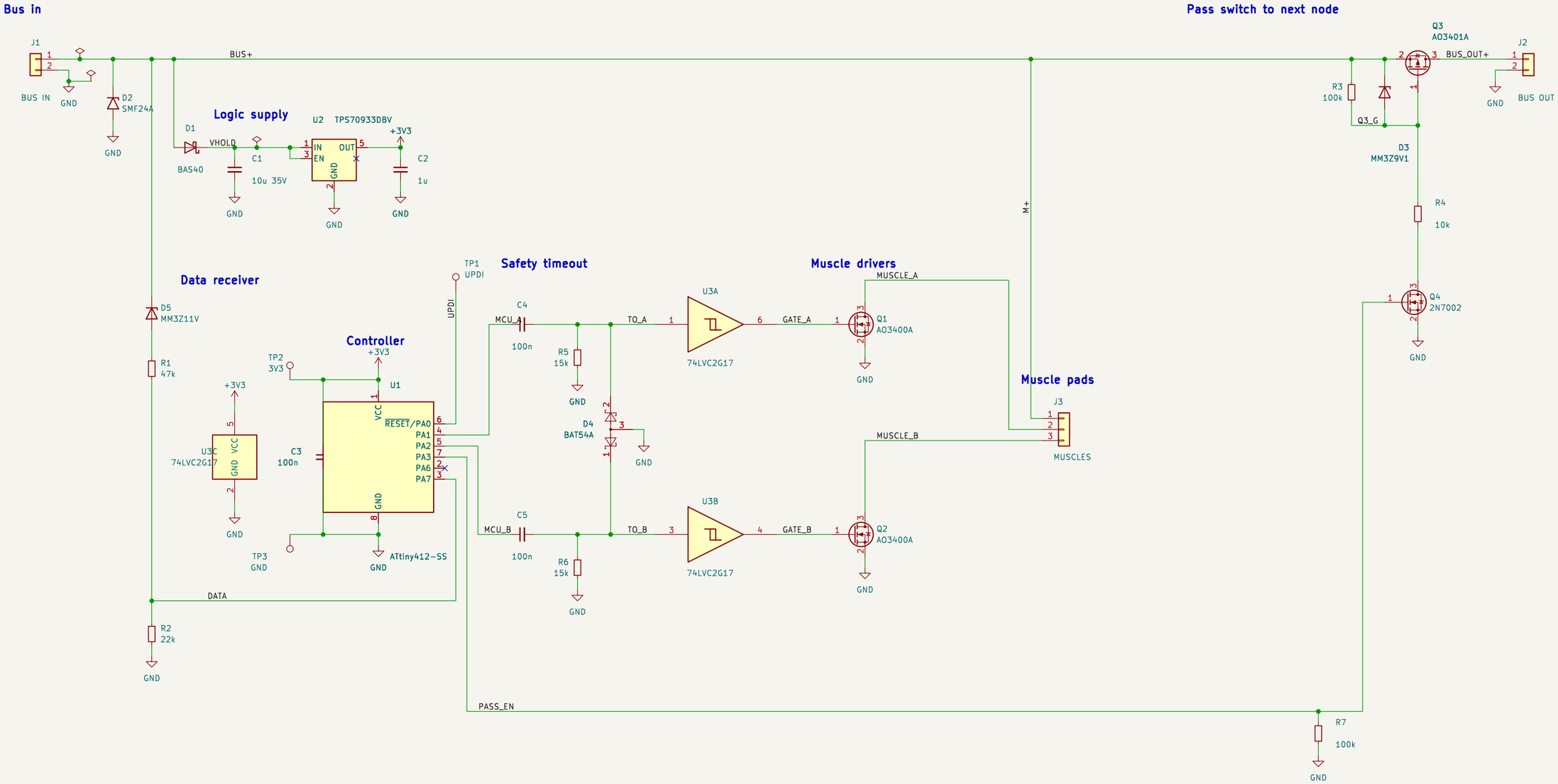


Notes

Flow: bus power and data enter at J1 (left) and leave at J2 (right) only when Q3 is switched on.

Bus: 20 V idle / data "1", about 12 V for data "0". D5 + R1/R2 shift it to ~2.9 V / ~0.3 V at PA7 (USART0 RXD).

PA1 / PA2 (TCA0 WO1 / WO2) drive muscles A / B through C4–R5 and C5–R6: each pulse ends by ~1.8 ms at the latest.

D4 clamps the coupling nodes on falling edges; the U3 Schmitt buffers give the FETs a clean turn-off.

PA3 (PASS_EN) switches Q4, which pulls Q3 on. R3 and R7 keep Q3 off at power-up; D3 limits Q3 VGS to 9.1 V.

J3 pad order: 1 = M+ (bus+, common end of both muscles), 2 = muscle A, 3 = muscle B.

J1, J2, J3 are bare wire pads: footprints still to be drawn. TP1–TP3 on the bottom side for a pogo-pin UPDI jig.

D2 is unidirectional: cathode (pin 1) to bus+. Sketch for review: check pinouts and ratings against datasheets.

Drawn as signal flow, left to right; not yet checked against datasheets
Option A node: ATtiny412, voltage-dip data, capacitor-coupled gate timeout

Sheet: /
File: sma_node.kicad_sch

Title: SMA limb node: two-wire bus

Size: A3	Date: 2026-10-08	**Rev: 0.2**
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